HH-I-585C
October 17, 1974
SUPERSEDING
Fed. Spec. HH-I-585B
March 3, 1969

FEDERAL SPECIFICATION

INSULATION, THERMAL (VERMICULITE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers expanded or exfoliated vermiculite thermal insulating material in the dry loose condition for use as fill insulation to restrict the flow of heat.

1.2 Classification.

1.2.1 Types and classes. Vermiculite loose fill insulation shall be of the following types and classes as specified (see 6.2):

Type I - Extra coarse.
Type II - Coarse.
Type III - Fine.
Type IV - Extra fine.

Class 1 - Not treated for water repellency.
Class 2 - Treated for water repellency.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge Polyolefin.
UU-S-48 - Sacks, Paper, Shipping.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 136 - Method of Test for Sieve or Screen Analysis of Fine and Coarse Aggregates.
C 177 - Method of Test for Thermal Conductivity of Materials by Means of the Guarded Hot Plate.
C 516 - Specification for Vermiculite Loose Fill Insulation.
C 520 - Method of Test for Density of Granular Loose Fill Insulations.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P St., N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Vermiculite loose fill insulation shall be produced by the expanding or exfoliating of natural vermiculite ore by grading and heating and shall meet the requirements of ASTM C 516. Class II shall be treated to produce a specific property such as water repellency.

3.1.1 Class 1. Class 1 material shall be a loose fill insulation composed of expanded or exfoliated vermiculite.

3.1.2 Class 2. Class 2 material shall be a loose fill insulation composed of expanded or exfoliated vermiculite treated for water repellency.

3.2 Water repellency, class 2. When tested in accordance with 4.4.5, the water repellency of class 2 vermiculite insulation shall be not less than 175 milliliters (ml.) of water repelled.

3.3 Combustibility. When tested in accordance with 4.4.6, vermiculite insulation shall not spark or ignite.

3.4 Workmanship. Vermiculite insulation shall be clean and free of extraneous particles.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

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4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each type and class of vermiculite insulation submitted for inspection at the same time.

4.2.2 Sampling for tests. Samples for tests shall be in accordance with level S-2, of MIL-S-105D. The acceptable quality level (AQL) shall be 2.5 percent defective for specimens used in each test.

4.3 Examination.

4.3.1 End item. Vermiculite shall be examined for defects listed in table I. The acceptable quality levels (AQL's) in MIL-STD-105 shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Types and class not as specified.	X	
Bags not fully closed.	X	
Bags split or torn.	X	
Holes in bags.	X	
Dirty or damp insulation.	X	
Marking omitted, incomplete, illegible, or in wrong location.		X

4.3.2 Inspection of preparation of delivery. An inspection shall be made to determine that the packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure.
Contents	Net weight is more or less than required.

4.4 Test methods.

4.4.1 Acceptable Quality Level. The AQL for tests shall be 4.0 percent defective.

4.4.2 Density. The density of vermiculite shall be determined in accordance with ASTM C 520, (see 3.1).

4.4.3 Thermal conductivity. The thermal conductivity of vermiculite shall be determined in accordance with ASTM C 177 (see 3.1).

4.4.4 Size. The size of vermiculite shall be determined in accordance with ASTM C 136 (see 3.1).

4.4.5 Water repellency, class 2. The water repellency of class 2 vermiculite insulation shall be determined in accordance with the following procedure (see 3.2).

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4.4.5.1 A rigid plastic tube 2 inch ID by 12 inch long with a 16 mesh plastic screen covering firmly fastened or adhered to the bottom opening shall be employed. The tube shall be marked at 7.77 inches from screen covered end indicating a volume of 400-ml. The plastic tube shall be filled with the vermiculite insulation to level slightly above the 400-ml. mark. It shall then be compacted by dropping the tube from a height of approximately 3 inches on a large rubber stopper for 10 drops. As the sample compacts to a level below the 400-ml. mark, additional material shall be added so that after the tenth drop the level of the sample is within 1/8 inch of the 400-ml. mark. The tube shall be kept in a vertical position during compaction.

4.4.5.2 With the tube supported in a vertical position and a beaker positioned under the tube, simultaneously start timing and rapidly pour 250 ml. of water onto the vermiculite insulation. Allow 3 minutes for the water to drain through specimen, then tilt tube at approximately 45° to drain water collected on screen.

4.4.5.3 Pour water collected in beaker into graduate and record amount repelled by the treated vermiculite insulation.

4.4.5.4 After removing the specimen, wash and dry the tube before the next test.

4.4.5.5 At least 3 tests shall be conducted using separate samples of the vermiculite insulation and the results averaged.

4.4.6 Combustibility. The vermiculite shall be held in a suitable fixture and positioned in direct contact with the flame of a Bunsen burner for 15 seconds (see 3.3).

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Loose fill vermiculite insulation shall be packed in quanitites as specified in the contract or purchase order in sacks conforming to UU-S-48, level A, or in plastic bags constructed from material conforming to L-P-378, a minimum of 4 mils thick. The plastic bags shall be heat sealed after filling. Excess air within the filled bag shall be kept to a practicable minimum. When the UU-S-48 sack is used, the gross weight shall not exceed 40 pounds.

5.1.2 Level B. Loose fill insulation shall be packed in quanitites as specified in the contract or purchase order in sacks conforming to UU-S-48, level B or in plastic bags constructed from material conforming to L-P-378, a minimum of 4 mils thick. The plastic bags shall be heat sealed after filling. Excess air within the filled bag shall be kept to a practicable minimum. When the UU-S-48 sack is used, the gross weight shall not exceed 40 pounds.

5.1.3 Level C. Vermiculite insulation shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification rules and the National Motor Freight Classification rules.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the bags and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the bags and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Intended for the insulation of structures designed for human occupancy in the temperature range of -50° to +180°F.

6.1.2 Type II. Intended for the insulation of structures designed for human occupancy in the temperature range of -60° to +180°F, for structures intended for cold storage or low temperature testing, and for structures or equipment operated in the temperature range of -100° to +2000°F.

6.1.3 Type III. Intended for insulation of structures or equipment operated in the temperature range from liquid air to +2000°F.

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6.1.4 Type IV. Intended for insulation of structures or equipment with large temperature differences across the insulation and extremes of temperature range described in 6.1.3.

6.1.5 Class 1. Intended for insulation in areas where condensed moisture is not a factor. May also be used as an absorbent material for air shipment of dangerous liquids packaged in glass containers, paint, etc., for mailing through the U.S. Postal Service.

6.1.6 Class 2. Intended for fill insulation of masonry walls where condensed moisture may be a factor.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of insulation (see 1.2).
(c) Size of lot, if different from 4.2.1.
(d) Selection of applicable level of packing required (see 5.1).

6.3 Description of vermiculite. Vermiculite is a micaceous mineral containing a small amount of combined water. The crude vermiculite ore is mined, cleaned, and milled to a controlled size. It is then heated to a temperature of about 2000°F which causes each granule to expand about 12 times its original size. The expanded vermiculite contains thousands of entrapped air cells which account for its thermal insulation and light weight.

6.4 Coverage. The approximate coverage of vermiculite insulation for masonry walls can be determined in accordance with table III.

TABLE III. Approximate coverage for masonry walls, per 4 cubic feet of masonry fill

Type of wall	Wall area, square feet
2 inch cavity.	24
2-1/2 inch cavity.	20
4-1/2 inch cavity.	11
6 inch block, 2 or 3 core.	21 (23-1/2 blocks)
8 inch block, 2 or 3 core.	14-1/2 (16-1/2 blocks)
12 inch block, 2 or 3 core.	8 (9 blocks)

MILITARY CUSTODIANS:

Navy - YD
Air Force - 84

User activities:

Navy - MC
Army - CE

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE-NBS
GSA-FSS, PCD
HUD-TCS
INTERIOR-BPA

Military Coordinating Activity:

Navy - YD

U. S. GOVERNMENT PRINTING OFFICE : 1975 - 585-647/1418

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HH-I-585B
March 3, 1969
SUPERSEDING
Int. Fed. Spec. HH-I-00585A(GSA-FSS)
July 11, 1967,
Fed. Spec. HH-I-585
October 12, 1956, and
Mil-V-15196B
May 3, 1962

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1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers expanded or exfoliated vermiculite thermal insulating material in the dry loose condition for use as fill insulation to restrict the flow of heat.

1.2 Classification.

1.2.1 Types and classes. Vermiculite loose fill insulation shall be of the following types and classes as specified (see 6.2):

Type I - Extra coarse.
Type II - Coarse.
Type III - Fine.
Type IV - Extra fine.

Class 1 - Not treated for water repellency.
Class 2 - Treated for water repellency.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-S-48 - Sacks, Paper, Shipping.
PPP-D-723 - Drums, Fiber.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian agencies).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Publications:

C 136 - Method of Test for Sieve or Screen Analysis of Fine and Coarse Aggregates.
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C 516 - Specification for Vermiculite Loose Fill Insulation.
C 520 - Method of Test for Density of Granular Loose Fill Insulations.

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National Classification Board:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board, 1616 P St., N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd. Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material.

3.1.1 Class 1. Class 1 material shall be a loose fill insulation composed of expanded or exfoliated vermiculite.

3.1.2 Class 2. Class 2 material shall be a loose fill insulation composed of expanded or exfoliated vermiculite treated for water repellency.

3.2 Density. When tested in accordance with 4.4.2, the density of vermiculite insulation shall conform with the requirements of table I.

TABLE I. Density

Size designation	Dry loose weight, lb per cu. ft.	
Type	Min.	Max.
I	3-1/2	7
II	4	8
III	5	9
IV	6	10

3.3 Thermal conductivity. When tested in accordance with 4.4.3, the thermal conductivity of vermiculite insulation shall conform with the requirements of table II.

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TABLE II. Thermal conductivity

Mean temperature	Thermal conductivity, max, BTU in/hr sq ft °F.		
°F	Type I	Types II and III	Type IV
0	0.44	0.44	0.44
75	0.50	0.50	0.50
250	0.75	0.63	0.63
500	-	-	0.85
750	-	-	1.10
1000	-	-	1.40
1200	-	-	1.60

3.4 Size. When tested in accordance with 4.4.4, the size of vermiculity insulation shall conform with the requirements of table III.

TABLE III. Size

Type	Accumulative percent retained on sieves having square holes, by weight													
	Sieve size													
	3/8 inch		No. 4		No. 8		No. 16		No. 30		No. 50		No. 100	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max
I	0	40	30	90	65	98	85	100	-	-	-	-	-	-
II	-	-	0	5	20	80	75	99	90	100	97	100	-	-
III	-	-	-	-	0	10	20	70	65	95	75	98	90	100
IV	-	-	-	-	-	-	0	5	15	65	60	98	90	100

3.5 Water repellency, class 2. When tested in accordance with 4.4.5, the water repellency of class 2 vermiculite insulation shall be not less than 175 milliliters (ml) of water repelled.

3.6 Combustibility. When tested in accordance with 4.4.6, vermiculite insulation shall not spark or ignite.

3.7 Workmanship. Vermiculite insulation shall be clean and free of extraneous particles.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each type and class of vermiculite insulation submitted for inspection at the same time.

4.2.2 Visual examination of end item. Samples for visual examination shall be selected in accordance with level S-3 in MIL-STD-105.

4.2.3 Samples for tests. The number of samples for tests shall be selected in accordance with ASTM C 516.

4.3 Examination.

4.3.1 End item. Vermiculite insulation selected in accordance with 4.2.2, shall be examined for defects listed in table IV. The Acceptable Quality Levels (AQL's) for defects shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE IV. Classification of defects, end item.

Defects	Major	Minor
Types and class not as specified.	X	
Dirt in insulation.	X	
Insulation not free of extraneous particles.	X	
Damage or defects not affecting function or serviceability.		X

4.3.2 Inspection of preparation of delivery. An inspection shall be made to determine that the packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table V. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE V. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure.
Contents	Net weight is more or less than required.

4.4 Test methods.

4.4.1 Acceptable Quality Level. The AQL for tests shall be 4.0 percent defective.

4.4.2 Density. The density of vermiculite shall be determined in accordance with ASTM C 520, (see 3.2).

4.4.3 Thermal conductivity. The thermal conductivity of vermiculite shall be determined in accordance with ASTM C 177 (see 3.3).

4.4.4 Size. The size of vermiculite shall be determined in accordance with ASTM C 136 (see 3.4).

4.4.5 Water repellency, class 2. The water repellency of class 2 vermiculite insulation shall be determined in accordance with the following procedure (see 3.5):

4.4.5.1 A rigid plastic tube 2 inch ID by 12 inches long with a 16 mesh plastic screen covering firmly fastened or adhered to the bottom opening shall be employed. The tube shall be marked at 7.77 inches from screen covered end indicating a volume of 400-ml. The plastic tube shall be filled with the vermiculite insulation to level slightly above the 400-ml. mark. It shall then be compacted by dropping the tube from a height of approximately 3 inches on a large rubber stopper for 10 drops. As the sample compacts to a level below the 400-ml. mark, additional material shall be added so that after the tenth drop the level of the sample is within 1/8 inch of the 400-ml. mark. The tube shall be kept in a vertical position during compaction.

4.4.5.2 With the tube supported in a vertical position and a beaker positioned under the tube, simultaneously start timing and rapidly pour 250 ml. of water onto the vermiculite insulation. Allow 3 minutes for the water to drain through specimen, then tilt tube at approximately 45° to drain water collected on screen.

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4.4.5.3 Pour water collected in beaker into graduate and record amount repelled by the treated vermiculite insulation.

4.4.5.4 After removing the specimen, wash and dry the tube before the next test.

4.4.5.5 At least 3 tests shall be conducted using separate samples of the vermiculite insulation and the results averaged.

4.4.6 Combustibility. The vermiculite shall be held in a suitable fixture and positioned in direct contact with the flame of a Bunsen burner for 15 seconds (see 3.6).

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Vermiculite insulation shall be packed in paper shipping sacks conforming to UU-S-48, type 1, 6-6X, or fiber drums conforming to PPP-D-723, type II or III.

5.1.2 Level B. Vermiculite insulation shall be packed as specified in 5.1.1, except the shipping sacks shall be type 1, 4-4X and the fiber drums type I, grade A.

5.1.3 Level C. Vermiculite insulation shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Unform Freight Classification or National Motor Freight Classification Rules.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Intended for the insulation of structures designed for human occupancy in the temperature range of -50° to +180°F.

6.1.2 Type II. Intended for the insulation of structures designed for human occupancy in the temperature range of -60° to +180°F, for structures intended for cold storage or low temperature testing, and for structures or equipment operated in the temperature range of -100° to +2000°F.

6.1.3 Type III. Intended for insulation of structures or equipment operated in the temperature range from liquid air to +2000°F.

6.1.4 Type IV. Intended for insulation of structures or equipment with large temperature differences across the insulation and extremes of temperature range described in 6.1.3.

6.1.5 Class 1. Intended for insulation in areas where condensed moisture is not a factor. May also be used for dunnage purposes, such as air shipment of dangerous liquids.

6.1.6 Class 2. Intended for fill insulation of masonry walls where condensed moisture may be a factor.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of insulation (see 1.2).
(c) Size of lot, if different from 4.2.1.
(d) Selection of applicable level of packing required (see 5.1).

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6.3 Description of vermiculite. Vermiculite is a micaceous mineral containing a small amount of combined water. The crude vermiculite ore is mined, cleaned, and milled to a controlled size. It is then heated to a temperature of about 2000°F which causes each granule to expand about 12 times its original size. The expanded vermiculite contains thousands of entrapped air cells which account for its thermal insulation and light weight.

6.4 Coverage. The approximate coverage of vermiculite insulation for masonry walls can be determined in accordance with table VI.

TABLE VI. Approximate coverage for masonry walls, per 4 cubic feet of masonry fill

Type of wall	Wall area, square feet
2 inch cavity	24
2-1/2 inch cavity	20
4-1/2 inch cavity	11
6 inch block, 2 or 3 core	21 (23-1/2 blocks)
8 inch block, 2 or 3 core	14-1/2 (16-1/2 blocks)
12 inch block, 2 or 3 core	8 (9 blocks)

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 84

Review activities:

Army - ME
Navy - OS
Air Force - 85

Military Coordinating Activity:

Army - ME

Preparing activity:

GSA-FSS

Order for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

HH-I-585B INSULATION, THERMAL (VERMICULITE).

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

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POSTAGE AND FEES PAID
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GENERAL SERVICES ADMINISTRATION

OFFICIAL BUSINESS

General Services Administration
Federal Supply Service (FMSH)
Washington, D. C. 20406

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